MPG No. 14.5.52 $\bar{4}2'm'$ (-42'm' setting) [Type III, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)}$$
$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{M}_z^{(2)}$$
$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ -M_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzy} = \mathbb{M}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xxxy} & M_{xxxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxxy} & -M_{xxxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 \\ 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{\beta(2)}$$
$$M_{yzzx} = -\mathbb{M}_{xy}^{(5)} - \mathbb{T}_{z}^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)}$$
$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{xxyx} & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_{z}^{\beta(1)} + \mathbb{T}_{z}^{\beta(2)}$$
$$M_{xxyx} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_{z}^{\beta(1)} - \mathbb{T}_{z}^{\beta(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} - \mathbb{T}_{z}^{\beta(2)}$$
$$M_{yzxz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{\beta(2)}$$
$$M_{xyxx} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{xyzz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 4\mathbb{T}_{z}^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & -M_{yzxz} & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)} \\ M_{yzxz} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)} \\ M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{\beta(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxxy} & 0 \\ M_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzxy} & 0 \\ M_{xxyx} & 0 & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{xyyy} & 0 \\ 0 & 0 & -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} + M_{zzxy} \end{bmatrix}$$

$$\begin{aligned} M_{xxxy} &= -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} \\ M_{zzxy} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} \\ M_{xxyx} &= -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} \\ M_{yzzx} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} \\ M_{xyyy} &= -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} \end{aligned}$$